Social Impact Project · Loita Ward, Narok County, Kenya

CLEAN WATER FOR 500 STUDENTS

Autonomous solar water supply & treatment system · Off-grid · 20-year lifespan

500	$5m^3$	2km	0€ OpEx
Students & Families	Daily Supply	Pipeline	100% Autonomous

THE PROBLEM

159 — **E.Coli MPN/100 ml**

WHO limit: 0. Severe fecal contamination confirmed by lab analysis. Direct cause of waterborne disease.

109 — **NTU Turbidity**

4× above the legal limit of 25 NTU. Water shared with Maasai livestock at the surface pond.

4 km — **Daily Walk to Water**

Students miss school hours collecting water. No grid electricity within 6 hours of the site.

BUDGET BREAKDOWN

Solar-Powered Pumping System	2,150€
Transmission Line	2,970€
Storage Tanks (3x5000L)	1,780€
Pretreatment (Aeration + Sedimentator + URF)	3,100€
Desinfection (SSF + Biochar)	3,960€
Civil Works & Logistics	5,680€
TOTAL (15% Margin)	22,500€

THE SOLUTION

1 **Solar Pumping** — Autonomous off-grid raw water extraction

2 **2km Piping** — HDPE pipeline overcoming 25m elevation gain

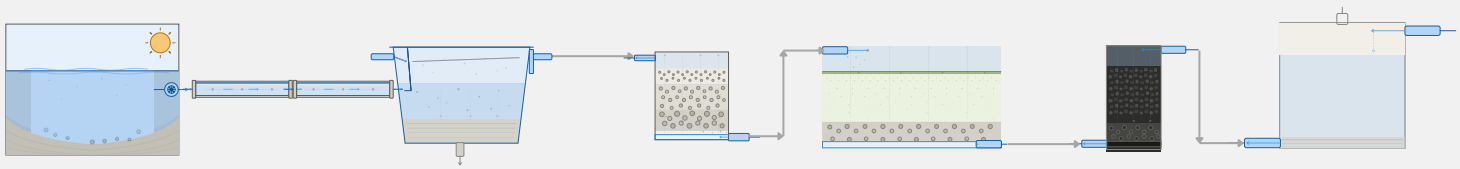
3 **Settling Tank** — For removing coarse particles

4 **Roughing Filter** — Fine solids adsorption in gravel bed

5 **Slow Sand Filter** — The Biological Filter for water desinfection

6 **Biochar Adsorption** — Innovative Technology for bad odor and taste

7 **Clean Water Tank** — Suppling over 500 students and families



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Join our Engineering & ESG Initiative

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100% transparent · All funds go to the field